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SafeRide - AI Enabled Smart Road Safety Violation Detection And Monitoring System Using Neural Vision

BE_DS_Group No: 05
Name of The Guide:
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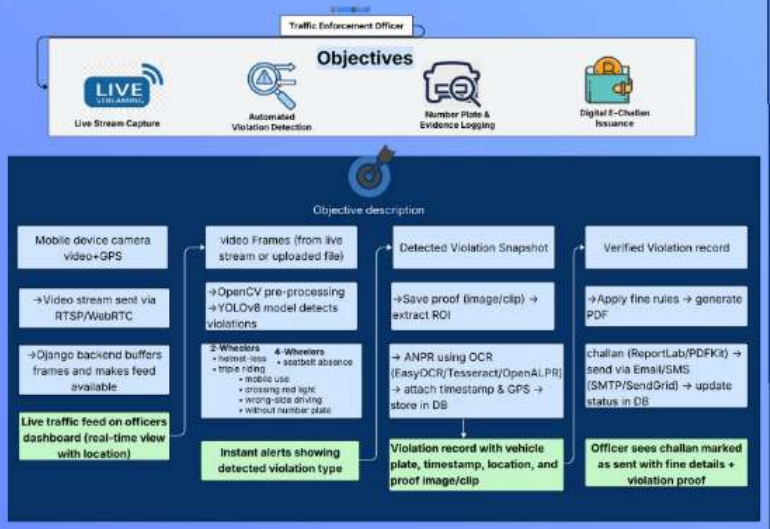
1. Problem Statement:

Do you really know how safe the roads around you are? With the rapid growth of vehicles and increasing traffic density, road safety has become a serious concern. Frequent traffic rule violations and unsafe driving behaviors continue to increase the risk of accidents. Limited monitoring and enforcement allow many violations to go unnoticed, putting countless lives at risk.

2. Objectives:

- Live Stream Capture
- Automated Violation Detection
- Number plate & Evidence Logging
- Digital E-Challan Issuance

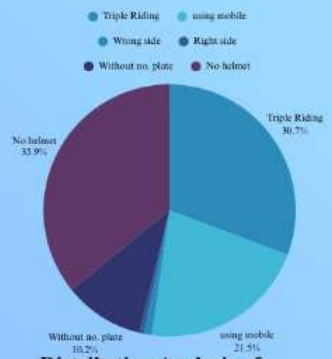
3. System Architecture:



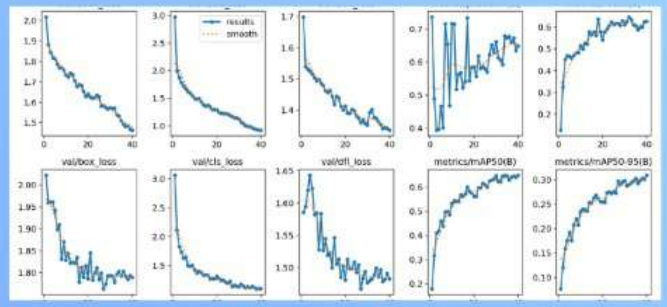
4. Results:

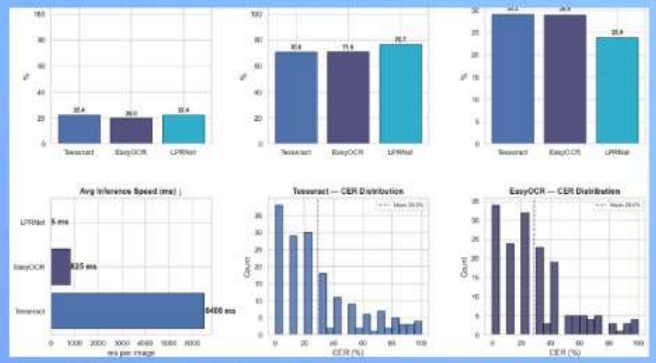
Real-time traffic violation detection using the YOLOv8 model

Accuracy of the LPRNet OCR Engine

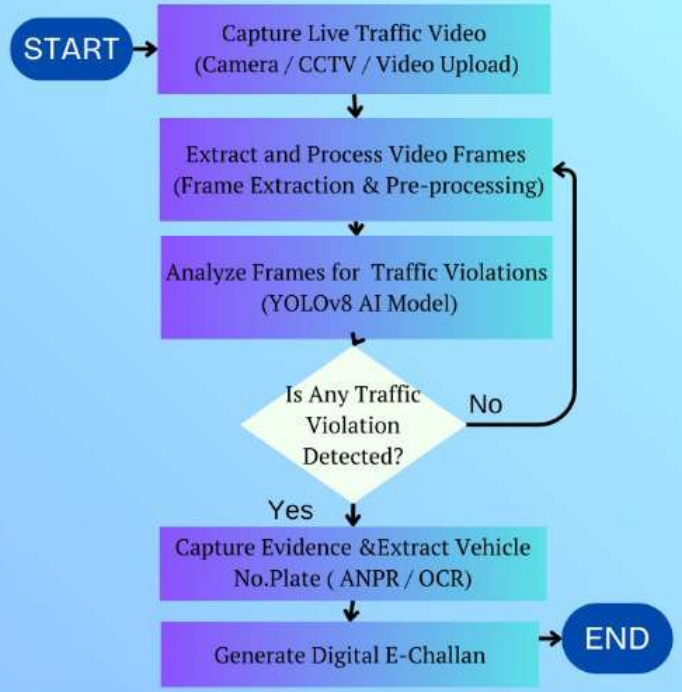


Distribution Analysis of Traffic Violations





5. Work Flow:



6. References:

- [1] Pa et al. (2025): Detects wrong-way driving using YOLOv9 and CNN for intelligent traffic monitoring.
- [2] Deshpande et al. (2024): Vision-based method for two-wheeler rider identification and triple-riding detection.
- [3] Said et al. (2024): AI-based system for helmet violation detection to improve road safety.

7. Conclusion:

SafeRide uses AI-powered vision to detect traffic violations in real time and automate enforcement. This intelligent system enhances road safety, efficiency, and smart traffic management.